

The goal is to test whether the **5 composition likelihood methods** differ in estimation accuracy across the sampled **K = 3 proportion simplex** and across the **972 factorial experiments**. The current output already gives three accuracy responses for each method: **RMSE, L1 norm, and L ∞ norm**, plus the true simplex point $\mathbf{t}$, $\mathbf{t}_1$, and $\mathbf{t}_2$. The script also already runs univariate Friedman tests, paired Wilcoxon tests, and mixed models separately for RMSE, L1, and L ∞ .

```
hake.R p1, p2, p3 mesh_id sim_id
```

The existing tests are useful, but they treat each accuracy metric separately. A stronger analysis should test whether methods differ in their **joint accuracy profile**, while also accounting for:

1. the paired structure of the 5 methods fit to the same simulated dataset,
2. repeated simulations within each simplex point,
3. repeated simplex points within each experiment,
4. factorial design effects across the 972 experiments, and
5. possible method differences that vary over the simplex.

The runner script places each experiment's , emmeans summaries, simplex samples, and plots into experiment-specific folders such as , etc., using the input stem , , and so on. hake.csv ex1 ex2 hake_ex1 hake_ex2

Use the design matrix as the **experiment-level metadata table**. It has one row per experiment and contains fields such as `example_id`, `inp_file`, `design_block`, `dist_code`, `G`, `theta_true`, `theta_CV`, `sigma`, `mean_nsamp`, `ln_sd`, `nb_size`, `Nmin`, `Nmax`, `N_range_label`

<https://chatgpt.com/c/69fcd70f-6fe8-83e8-8a80-01148703e365>

The is important. It identifies the same simulated dataset evaluated by all 5 likelihood methods, so method comparisons should be treated as **paired**. dataset_id

Then reshape to long format:

```
acc_long <- all_wide %>%
  pivot_longer(
    cols = matches("^~(rmse|l1_norm|l1nf_norm)_(i|ii|iii|iv|v)$"),
    names_to = c("metric", "method"),
    names_pattern = "(rmse|l1_norm|l1nf_norm)_(i|ii|iii|iv|v)",
    values_to = "accuracy"
  ) %>%
  mutate(
    method = factor(method, levels = c("i", "ii", "iii", "iv", "v")),
    metric = factor(metric, levels = c("rmse", "l1_norm", "l1nf_norm"))
  )
```

2. Approach A: Repeated-measures MANOVA on RMSE, L1, and L_∞

This is the most direct multivariate test. The response vector is:

$$\mathbf{y}_m = (\text{RMSE}_m, L1_m, L_\infty_m)$$

for method $m = i, \dots, v$.

A practical setup is to aggregate over simulations at each experiment $\times$ simplex point $\times$ method combination:

```
acc_mv <- acc_long %>%
  group_by(
    example_id, mesh_id, method,
    p1, p2, p3,
    design_block, G, theta_true, theta_cv, sigma,
    mean_nsamp, ln_sd, nb_size, N_range_label
  ) %>%
  summarize(
    rmse = mean(accuracy[metric == "rmse"], na.rm = TRUE),
    L1 = mean(accuracy[metric == "l1_norm"], na.rm = TRUE),
    L1nf = mean(accuracy[metric == "l1nf_norm"], na.rm = TRUE),
    .groups = "drop"
  )
```

Then fit a MANOVA:

```
fit_manova <- manova(
  cbind(rmse, L1, L1nf) ~ method * design_block +
    method * G +
    method * theta_true +
    method * theta_cv +
    method * sigma +
    method * mean_nsamp +
    method * p1 + method * p2,
  data = acc_mv
)

summary(fit_manova, test = "Pillai")
summary.aov(fit_manova)
```

Use **Pillai's trace** as the default test because it is generally more robust than Wilks' lambda when covariance assumptions are imperfect.

Interpretation:

This tests whether the 5 methods differ in their joint accuracy vector and whether those differences depend on factorial design settings or simplex location.

3. Approach B: Multivariate mixed-effects model using metric as a response dimension

This is probably the most flexible analysis. Stack the three metrics into one long response and model as a response dimension. This allows method differences to vary by metric, experiment, and simplex point. `metric`

```
library(lme4)
library(lmerTest)
library(emmeans)

fit_mv_lmm <- lmer(
  log1p(accuracy) ~ method * metric +
    method * design_block +
    method * G +
    method * theta_true +
    method * theta_CV +
    method * sigma +
    method * mean_nsamp +
    method * p1 + method * p2 +
    (1 | example_id) +
    (1 | simplex_id) +
    (1 | dataset_id),
  data = acc_long
)

anova(fit_mv_lmm)

emmeans(fit_mv_lmm, pairwise ~ method | metric, adjust = "holm")
emmeans(fit_mv_lmm, pairwise ~ method, adjust = "holm")
```

This model tests:

$$\log(1 + \text{accuracy}) \sim \text{method} \times \text{metric}$$

with random effects for experiment, simplex point, and simulated dataset.

Interpretation:

This estimates whether methods differ overall, whether method differences depend on the accuracy metric, and whether those differences persist after accounting for experiment-level and simplex-level variation.

This is a natural extension of the current approach already used in , but it uses all three accuracy metrics jointly rather than one at a time. `lmer(rmse ~ method + (1 | id)) hake.R`

4. Approach C: Pairwise multivariate contrasts among methods

For each simulated dataset, compute paired differences between methods:

$$D_{from} = (\text{RMSE}_a - \text{RMSE}_{b,L} \quad 1_a - L \quad 1_b, L \quad \infty_a - L \quad \infty_b)$$

Then test whether the mean vector differs from zero.

```
make_pairwise_diff <- function(dat, m1, m2) {
  dat %>%
    select(example_id, mesh_id, sim_id, dataset_id,
```

```

    p1, p2, p3,
    design_block, G, theta_true, theta_CV, sigma,
    mean_nsamp, ln_sd, nb_size, N_range_label,
    method, metric, accuracy) %>%
  filter(method %in% c(m1, m2)) %>%
  pivot_wider(names_from = method, values_from = accuracy) %>%
  mutate(
    comparison = paste0(m1, "_minus_", m2),
    diff = .data[[m1]] - .data[[m2]]
  ) %>%
  select(-all_of(c(m1, m2)))
}

diff_i_iv <- make_pairwise_diff(acc_long, "i", "iv")

diff_i_iv_mv <- diff_i_iv %>%
  pivot_wider(names_from = metric, values_from = diff) %>%
  rename(L1 = L1_norm, Linf = Linf_norm)

fit_hotelling <- manova(cbind(rmse, L1, Linf) ~ 1, data = diff_i_iv_mv)
summary(fit_hotelling, test = "Pillai")

```

For factorial effects on the pairwise differences:

```

fit_diff_manova <- manova(
  cbind(rmse, L1, Linf) ~ design_block + G + theta_true +
    theta_CV + sigma + mean_nsamp + pi + p2,
  data = diff_i_iv_mv
)

summary(fit_diff_manova, test = "Pillai")

```

Interpretation:

This directly answers: "Does method i significantly outperform method iv across the simplex and factorial design?" Repeat for all 10 method pairs.

5. Approach D: PERMANOVA using the 3-dimensional accuracy vector

A nonparametric alternative is PERMANOVA. This is useful if RMSE, L1, and L_∞ are skewed or not close to multivariate normal.

```

library(vegan)

acc_mv2 <- acc_long %>%
  group_by(example_id, mesh_id, sim_id, method,
    p1, p2, p3, design_block, G, theta_true,
    theta_CV, sigma, mean_nsamp, ln_sd, nb_size,
    N_range_label) %>%
  summarize(
    rmse = accuracy[metric == "rmse"][1],
    L1 = accuracy[metric == "L1_norm"][1],
    Linf = accuracy[metric == "Linf_norm"][1],
    .groups = "drop"
  )

Y <- acc_mv2 %>% select(rmse, L1, Linf)

fit_perm <- adonis2(
  Y ~ method * design_block +

```

```

method * G +
method * theta_true +
method * theta_CV +
method * sigma +
method * mean_nsamp +
method * p1 + method * p2,
data = acc_mv2,
permutations = 999,
method = "euclidean"
)

fit_perm

```

For a paired version, use restricted permutations within where possible: dataset_id

```

fit_perm_pair <- adonis2(
  Y ~ method,
  data = acc_mv2,
  permutations = how(blocks = acc_mv2$dataset_id),
  method = "euclidean"
)

```

Interpretation:

PERMANOVA tests whether the multivariate cloud of accuracy values differs among methods. It is robust and easy to explain, but it does not provide estimates as directly as mixed models.

6. Approach E: Functional or spatial analysis over the simplex

Because the true proportions lie on a simplex, method performance may vary by location. For $K = 3$, the simplex can be represented by and, with $p_1 + p_2 + p_3 = 1$.

A useful analysis is a method-specific smooth surface:

```

library(mgcv)

fit_gam <- gam(
  log1p(accuracy) ~ metric * method +
    s(p1, p2, by = method, k = 30) +
    design_block + G + theta_true + theta_CV + sigma + mean_nsamp,
  data = acc_long,
  method = "REML"
)

summary(fit_gam)
anova(fit_gam)

```

You can also model method differences directly. For example, method v minus method iv:

```

diff_v_iv <- make_pairwise_diff(acc_long, "v", "iv")

fit_diff_gam <- gam(
  log1p(abs(diff)) ~ metric +
    s(p1, p2, by = metric, k = 30) +
    design_block + G + theta_true + theta_CV + sigma + mean_nsamp,
  data = diff_v_iv,
  method = "REML"
)

```

```
)

summary(fit_diff_gam)
```

Interpretation:

This tests whether one method is better mainly near the center of the simplex, near the edges, or near corners where one composition category is rare.

This approach aligns well with the current ternary plotting logic in , which already aggregates mean RMSE, L1, and L_∞ by and plots them over the simplex. `hake.R mesh_id`

7. Approach F: PCA of accuracy profiles

Build a 15-dimensional accuracy vector for each experiment $\times$ simplex point $\times$ simulation:

$$(\text{RMSE}_i, \dots, \text{RMSE}_v, L_{1_i}, \dots, L_{1_v}, L_{\infty_i}, \dots, L_{\infty_v})$$

Then use PCA to summarize dominant patterns.

```
acc_profile <- all_wide %>%
  select(
    example_id, mesh_id, sim_id, dataset_id,
    p1, p2, p3, design_block, G, theta_true, theta_CV,
    sigma, mean_nsamp, ln_sd, nb_size, N_range_label,
    matches("^(rmse|L1_norm|Linf_norm)_(i|ii|iii|iv|v)$")
  )

Y <- acc_profile %>%
  select(matches("^(rmse|L1_norm|Linf_norm)_(i|ii|iii|iv|v)$")) %>%
  mutate(across(everything(), log1p))

pca <- prcomp(Y, center = TRUE, scale. = TRUE)

pc_dat <- bind_cols(
  acc_profile %>% select(example_id, mesh_id, sim_id, p1, p2, p3,
    design_block, G, theta_true, theta_CV,
    sigma, mean_nsamp, N_range_label),
  as.data.frame(pca$x[, 1:5])
)

fit_pc1 <- lm(PC1 ~ design_block + G + theta_true + theta_CV +
  sigma + mean_nsamp + p1 + p2,
  data = pc_dat)

anova(fit_pc1)
summary(fit_pc1)
```

Interpretation:

PCA can show whether the main pattern is an overall accuracy gradient, a multinomial-versus-DM contrast, or a contrast among DM parameterizations.

8. Approach G: Rank-based dominance analysis

If the main question is “which method is best most often?”, rank the 5 methods within each dataset and metric.

```
rank_dat <- acc_long %>%
  group_by(dataset_id, metric) %>%
```

```
mutate(rank = rank(accuracy, ties.method = "average")) %>%
ungroup()

rank_summary <- rank_dat %>%
  group_by(metric, method) %>%
  summarize(
    mean_rank = mean(rank, na.rm = TRUE),
    prob_best = mean(rank == 1, na.rm = TRUE),
    .groups = "drop"
  )

rank_summary
```

Then model ranks:

```
fit_rank <- lmer(
  rank ~ method * metric +
    design_block + G + theta_true + theta_CV + sigma + mean_nsamp +
    p1 + p2 +
    (1 | example_id) + (1 | dataset_id),
  data = rank_dat
)

anova(fit_rank)
emmeans(fit_rank, pairwise ~ method | metric, adjust = "holm")
```

Interpretation:

This gives an intuitive result: the probability that each method is best, second-best, and worst across the factorial design and simplex.

9. Approach H: Multivariate analysis of the actual estimation error vector

Instead of reducing each estimate to RMSE, L1, or L ∞ , analyze the estimated composition error:

$$\hat{\mathbf{p}}_m - \mathbf{p}_{true}$$

For K = 3, the three errors sum to zero, so use two independent coordinates, such as err_1 and err_2

```
err_long <- all_wide %>%
  pivot_longer(
    cols = matches("p_hat_(i|ii|iii|iv|v)_([123])$"),
    names_to = c("method", "cat"),
    names_pattern = "p_hat_(i|ii|iii|iv|v)_([123])",
    values_to = "p_hat"
  ) %>%
  mutate(
    cat = as.integer(cat),
    p_true = case_when(
      cat == 1 ~ p1,
      cat == 2 ~ p2,
      cat == 3 ~ p3
    ),
    error = p_hat - p_true
  )

err_wide <- err_long %>%
  select(example_id, mesh_id, sim_id, dataset_id,
    p1, p2, p3, design_block, G, theta_true, theta_CV,
    sigma, mean_nsamp, method, cat, error) %>%
  pivot_wider(names_from = cat, values_from = error,
    names_prefix = "err")
```

```
fit_err_manova <- manova(
  cbind(err1, err2) ~ method * design_block +
    method * G +
    method * theta_true +
    method * theta_CV +
    method * sigma +
    method * mean_nsamp +
    method * p1 + method * p2,
  data = err_wide
)

summary(fit_err_manova, test = "Pillai")
```

Interpretation:
This tests whether methods differ in **bias direction**, not only in scalar accuracy. It can reveal whether one method tends to overestimate rare categories or pull estimates toward the simplex center.

Recommended analysis sequence

I would use four complementary analyses.

Step	Analysis	Main purpose
1	Multivariate mixed model	Primary test of method differences across RMSE, L1, and L ∞
2	Pairwise multivariate contrasts	Direct method-vs-method comparisons
3	GAM over simplex	Identify where methods differ across simplex location
4	Rank dominance analysis	Provide a clear summary of which method performs best most often

The best primary model is:

```
log1p(accuracy) ~ method * metric +
  method * design_block +
  method * G +
  method * theta_true +
  method * theta_CV +
  method * sigma +
  method * mean_nsamp +
  method * p1 + method * p2 +
  (1 | example_id) +
  (1 | simplex_id) +
  (1 | dataset_id)
```

This model respects the paired design, uses all three accuracy metrics, includes the 972-experiment design matrix, and allows method performance to vary across the sampled proportion simplex.